

AI Entrance Training Course

Ethics, Challenges, and Risks of Artificial Intelligence

Session 8 | Understanding Risks, Accountability, and Principles of Safe and Conscious Use

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Previous Session Review & Today's Objectives

Session 7 Recap: Data, The True Fuel of AI

Exploring the importance of data quality over quantity, the concept of overfitting, and basic principles of data cleaning.

Learning Objectives for This Session:

Algorithmic Bias: Understanding why and how unfair decisions occur in machines

Fake Content and Deepfakes: Recognizing the risks of face and voice forgery and detection methods

Future of Jobs and Regulations: A realistic look at the job market, cybersecurity, and legal liability

Responsible Usage: Learning a practical checklist for ethical and safe use of smart tools

Necessity of an Ethical Framework

Why Is Ethics in AI Important?

1. Direct Impact on Human Lives and Destinies

AI integration into critical decisions such as loan approvals, hiring, medical diagnosis, and judicial sentencing

2. Scale and Speed of Error Propagation

A human mistake harms only one person, but a flawed algorithm can impact millions of people in a single second

3. AI Lacks Conscience and Moral Understanding

Machines do not comprehend good and evil; they only execute mathematical patterns and require a human moral compass

What Is Algorithmic Bias?

Definition in Simple Terms:

The systematic and repeatable tendency of an AI system to produce unfair outcomes favoring or disadvantaging a particular group.

The Myth of Machine Neutrality: Contrary to popular belief, computer code is not inherently neutral and often reproduces hidden biases

A Mirror of the Past: AI learns societal patterns; if past discrimination existed, the model learns it as a rule

Hidden Behind Mathematics: Detecting these biases is challenging because they appear disguised as scientific numbers and probabilities

Where Do Errors Originate?

Three Main Roots of Bias in Intelligent Systems

1. Imbalanced Training Data (Data Bias)

When a demographic, linguistic, or cultural group is underrepresented in the initial training data

2. Model Design and Weighting (Algorithmic Bias)

Engineers setting flawed optimization metrics (e.g., maximizing clicks on social media without regard to news veracity)

3. Human Decisions and Labeling (Human Bias)

Subconscious transfer of personal preferences, prejudices, and viewpoints of the development team into program logic

Real-World Examples of Algorithmic Bias

1. Facial Recognition System Errors

Over 99% accuracy for lighter skin tones versus over 30% error rates for darker skin tones due to flawed initial training photos

2. Gender Stereotypes in Image Generation and Translation

Displaying images of men when searching for 'CEO or Engineer' and images of women for 'Nurse or Secretary'

3. Banking and Insurance Credit Scoring Systems

Rejection of loan applications from qualified individuals by algorithms purely due to residing in a specific neighborhood or postal code

The Fine Line Between Reality and Illusion

What Is a Deepfake and What Are Its Dangers?



1. Definition of Deepfake

Hyper-realistic synthesis of human faces, lip movements, and voices in video or audio using deep learning

2. Financial and Reputational Risks and Exploitation

Phone scams impersonating voices of family members or company executives, and fabricated videos to damage reputations

3. Countermeasures:

Establishing a 'family codeword' for emergency calls and verifying the primary source of sensational videos before believing them

Misinformation and Fake Content in the Age of AI

1. AI Hallucination Phenomenon

Chatbots fabricating completely fictitious statistics, historical events, or scientific citations with supreme confidence

2. Mass Production of Fake News

Generating hundreds of automated websites, fake accounts, and staged news photos in minutes to manipulate public opinion

3. The Need for 'Modern Media Literacy'

In today's world, 'seeing with your own eyes' is no longer proof of authenticity; verifying credible sources is essential

AI Security Risks in Simple Terms



1. Smart Phishing and Fraud

Crafting highly personalized, flawless emails and messages to steal user credentials without traditional typos

2. Corporate Confidential Data Leakage

Risks of exposing source code, contracts, or financial data when pasting proprietary info into public chatbots

3. Prompt Injection Attacks

Manipulating customer support chatbots with clever prompts to bypass security safeguards and extract internal data

The Impact of AI on Jobs: A Realistic Perspective



1. Automation of Repetitive Tasks (Not Job Elimination)

Routine tasks like basic data entry, simple translation, and summarizing are delegated to machines

2. Transformation of Roles and Productivity Boost

Professionals becoming 'managers of intelligent tools', shifting focus toward creativity, empathy, and strategic judgment

3. Emergence of Entirely New Career Paths

Surging demand for automation specialists, AI ethics auditors, prompt engineers, and model trainers

Accountability: When AI Makes a Mistake, Who Is at Fault?

1. A Challenging Example: Autonomous Vehicle Crash or Medical Error

Does the fault lie with the programmer, the vehicle manufacturer, the data provider, or the passenger?

2. The 'Human-in-the-Loop' Principle

AI is strictly an 'advisor and assistant'; the final sign-off in medical, financial, and legal matters remains with humans

3. The Shared Responsibility Framework

Software cannot be fined or imprisoned; deploying organizations and end users are ultimately responsible for reviewing outputs

Global Regulations

AI Governance and Global Regulations

1. Why Do We Need New Laws?

Balancing 'fostering innovation and scientific progress' with 'protecting citizen rights and societal safety'

2. Global Benchmark: The EU AI Act

Categorizing systems by risk level; outright bans on cognitive manipulation and strict oversight on medical/judicial uses

3. Mandatory Synthetic Content Watermarking

Mandating digital markers on AI-generated images, audio, and text to ensure full transparency for audiences

Additional Ethical Dimensions

Other Critical Ethical Challenges in AI

1. Copyright and Intellectual Property:

Using millions of artists' works and authors' books across the web to train models without consent or royalties

2. Transparency and the Black Box Dilemma:

The fundamental human right to know 'why and on what grounds' an algorithm made a specific decision about them

3. Emotional Dependence and Cognitive Decline:

The risk of intellectual passivity among learners and substituting authentic human relationships with chatbot companionship

4. High Energy and Water Consumption (Environment):

Immense power requirements of massive AI data centers and substantial water consumption for cooling compute servers

How Can We Use Artificial Intelligence Responsibly?

Audience	Key Action 1	Key Action 2
Everyday Users & Students	Fact-check critical statistics and scientific claims	Use as a learning assistant, not for plagiarism or cheating
Content Creators & Media	Maintain transparency with audience regarding AI content	Respect copyright and avoid voice cloning without consent
Managers & Businesses	Ensure rigorous protection of customer private data	Retain ultimate human oversight on hiring and financial decisions

Thought Challenge & Class Discussion

Interactive Activity: What If You Were the Decision-Maker?

Ethical Scenario for Discussion:

“A hospital purchased an AI system that diagnoses diseases with 95% accuracy, but makes errors in 5% of cases and no one can explain why it fails (black box). Should it be used?”

Follow-up Question: What conditions or policies would you establish to save more patient lives while managing the risk of error?

Final Review

Session 8 Summary

1. Awareness Over Fear: Understanding challenges like bias and deepfakes empowers us to become resilient, intelligent users.

2. Constant Human Oversight: AI is an assistive tool; ethical and legal responsibility for decisions always rests with humans.

3. Evolution of Careers: AI automates repetitive tasks but creates tremendous opportunities for those proficient in modern tools.

4. Three Core Principles of Responsible Use: Fact-checking information, safeguarding privacy, and maintaining honesty and transparency in content creation.

Addressing Concerns

Frequently Asked Questions About AI Ethics and Risks

Will AI rebel against humanity like in sci-fi movies?

No. Current models lack self-awareness, volition, and emotion; the real danger is human misuse, negligence, or over-reliance on these tools.

How can we detect a deepfake video or synthesized image?

Look closely at blinking patterns, lip-sync alignment, finger anatomy, background text glitches, and most importantly, verify the publisher source.

Is using AI-generated text or images legally prohibited?

General use is fully permitted; only impersonating real individuals, committing fraud, or infringing registered copyrights is subject to legal prosecution.

Observational & Analytical Exercise

Session 8 Assignment: Become an AI Ethics Auditor!

This Week's Mission: Spotting Bias or Fact-Checking AI

Complete one of the two experiments below and document your findings in a few concise lines.

Option 1 (Bias Audit): Ask an image generator to depict a 'successful doctor' or 'a family having lunch'; analyze the default demographic, cultural, or gender patterns.

Option 2 (Hallucination Hunt): Inquire about a niche local topic or an obscure publication from a chatbot and assess whether it fabricates fictional details.

Conclusion: Share your proposed solution for mitigating that flaw in the course discussion group.

End of Session 8 | AI Entrance Course

Thank You for Your Engagement and Participation!

Next Session: Course Wrap-up, Continuing Learning Roadmap, and Final Project

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